

Used Cars Prices Prediction with Machine Learning

A data science project



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Problem & Goal

- *Used cars prices vary widely depending on factors such as brand, mileage, and features, making valuation inconsistent.*
- *The goal is to build a machine learning model to accurately predict the resale price of a used car.*

Data & Features

Dataset Overview:

- 400,000+ used car records
- Source: hatla2ee.com (2024)

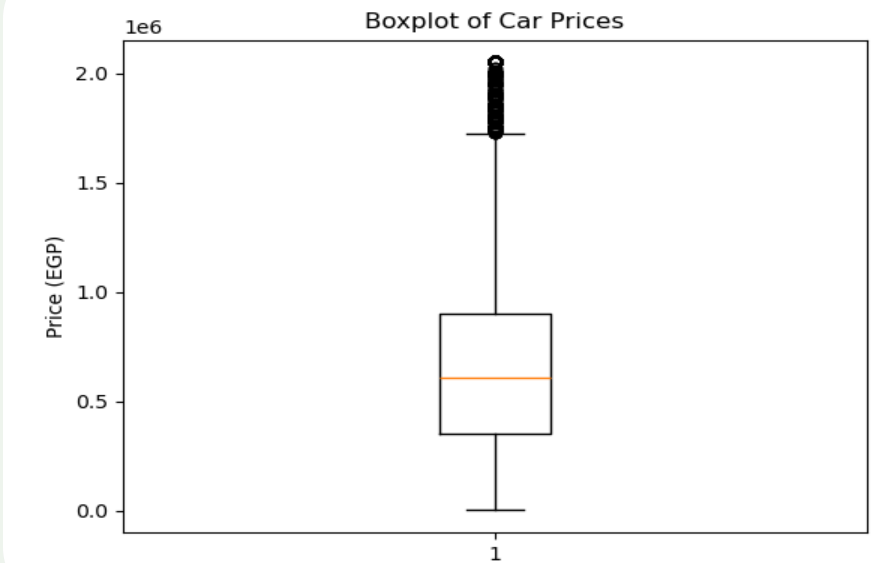
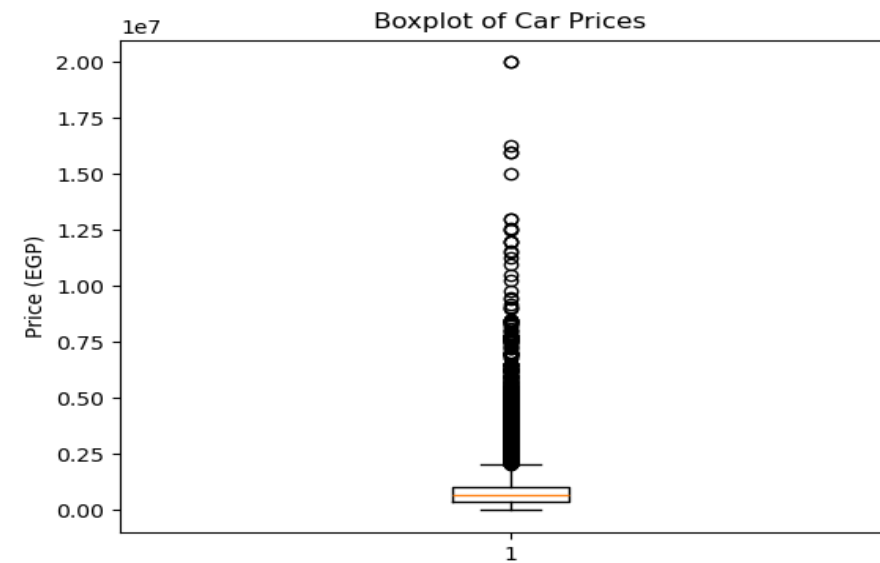
Key Features:

- Mileage
- Price
- Air conditioner, power steering, remote control
- Brand & Model

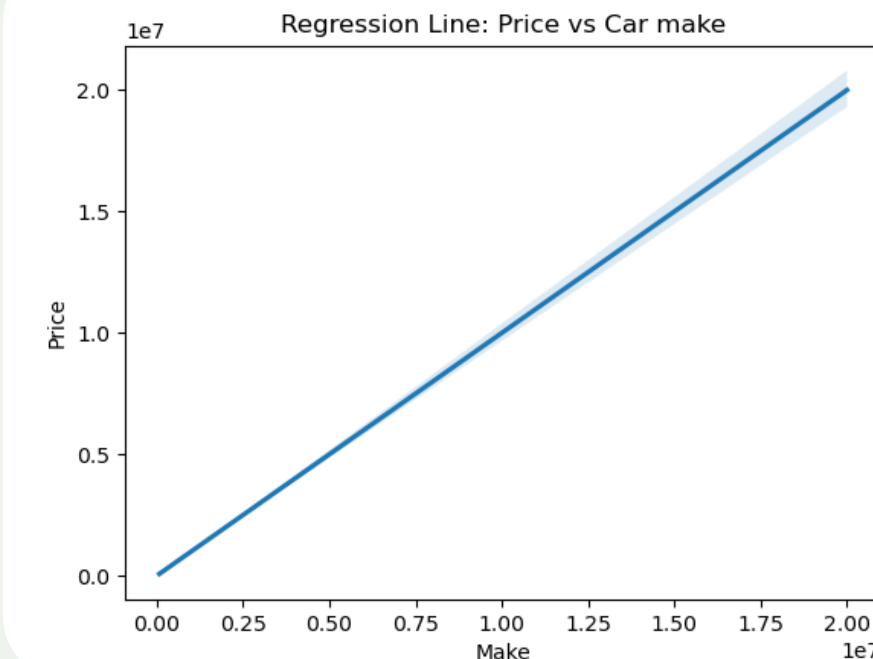
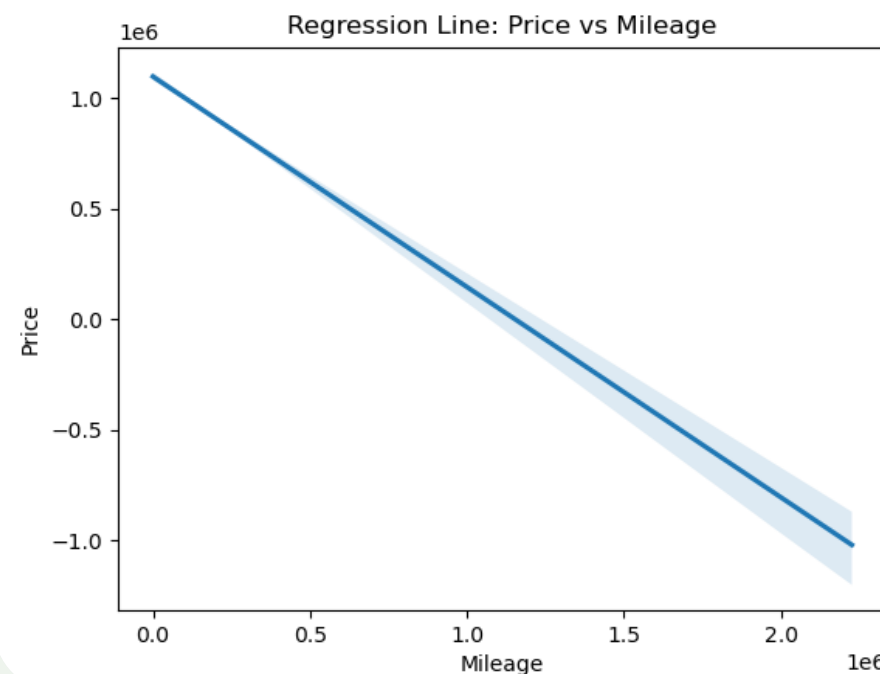
	Price	Color	Mileage	Make	Model	City	Date Displayed	Automatic Transmisior	Air Conditione	Power Steering	Remote Control	Item URL
2	1200000	Petroleum	131000	Fiat	Tipo	Kafr el-Dai	2/16/2024	1	1	1	1	https://eg.hatla2ee.com/en/car/fiat/tipo/6286701
3	2050000	Black	125000	Hyundai	Tucson Tu	Nasr city	2/16/2024	1	1	1	1	https://eg.hatla2ee.com/en/car/hyundai/Tucson-Turbo/6286695
4	2050000	Gray	84000	BMW	X1	Cairo	2/16/2024	1	1	1	1	https://eg.hatla2ee.com/en/car/bmw/x1/6254123
5	700000	Blue	75000	Chevrolet	Optra	Obour City	2/16/2024	1	1	1	1	https://eg.hatla2ee.com/en/car/chevrolet/optra/6286679
6	700000	Silver	25000	Chery	Arrizo 5	Obour City	2/16/2024	0	0	0	0	https://eg.hatla2ee.com/en/car/chery/Arrizo-5/6286673
7	1950000	Blue	33000	Kia	Sportage	Giza	2/16/2024	1	1	1	1	https://eg.hatla2ee.com/en/car/kia/sportage/6262769
8	400000	Black	238000	Hyundai	Verna	Faiyum	2/16/2024	0	1	1	1	https://eg.hatla2ee.com/en/car/hyundai/verna/6286602
9	870000	Brown	99000	Ford	Focus	El Haram	2/16/2024	1	1	1	1	https://eg.hatla2ee.com/en/car/ford/focus/6286581
10	575000	Silver	262000	Nissan	Tiida	Maadi	2/16/2024	0	0	0	0	https://eg.hatla2ee.com/en/car/nissan/tiida/6286779
11	640000	Silver	180000	Mitsubishi	Lancer Pur	Tagamo3 -	2/16/2024	1	0	0	0	https://eg.hatla2ee.com/en/car/mitsubishi/Lancer-Puma/6286767
12	1900000	Black	98000	Opel	Grandland	Nasr city	2/16/2024	1	1	1	1	https://eg.hatla2ee.com/en/car/opel/Grandland/6280913
13	830000	Red	67000	Nissan	Sunny	Alexandria	2/16/2024	1	1	1	1	https://eg.hatla2ee.com/en/car/nissan/sunny/6283629
14	150000	White	180000	Fiat	UO	Alexandria	2/16/2024	0	0	0	0	https://eg.hatla2ee.com/en/car/fiat/u0/6286515
15	325000	Gray	253000	Kia	Spectra	Al Shorouk	2/16/2024	1	0	0	0	https://eg.hatla2ee.com/en/car/kia/spectra/6286758
16	650000	Black	199000	Toyota	Corolla	Alexandria	2/16/2024	0	0	0	0	https://eg.hatla2ee.com/en/car/toyota/corolla/6286755
17	900000	White	240000	Jac	J3	6-Oct	2/16/2024	1	0	0	0	https://eg.hatla2ee.com/en/car/jac/j3/6286737
18	1170000	Black	48000	MG		5 Kafr el-Dai	2/16/2024	1	1	1	1	https://eg.hatla2ee.com/en/car/moris-garage/mg-5/6286712
19	285000	White	175000	Hyundai	Verna	Alexandria	2/16/2024	0	0	1	1	https://eg.hatla2ee.com/en/car/hyundai/verna/6286706
20	1500000	Black	82500	Toyota	Corolla	Cairo	2/16/2024	1	1	1	1	https://eg.hatla2ee.com/en/car/toyota/corolla/6279740
21	650000	White	150000	Nissan	Sunny	Alexandria	2/16/2024	1	1	1	1	https://eg.hatla2ee.com/en/car/nissan/sunny/6286692
22	90000	Red	85000	Chevrolet	Optra	Giza	2/16/2024	1	0	0	0	https://eg.hatla2ee.com/en/car/chevrolet/optra/6286685
23	430000	Black	200000	Skoda	Octavia A4	Alexandria	2/16/2024	1	0	0	0	https://eg.hatla2ee.com/en/car/skoda/Octavia-AA/6274731
24	280000	Dark blue	198000	Daewoo	Nubira 2	Giza	2/16/2024	1	1	1	1	https://eg.hatla2ee.com/en/car/daewoo/Nubira-2/6274769
25	240000	Silver	250000	Kia	Sephia	Tagamo3 -	2/16/2024	0	1	0	1	https://eg.hatla2ee.com/en/car/kia/sephia/6286683
26	300000	Red	37000	Changan	Benni	Helwan	2/16/2024	1	1	0	0	https://eg.hatla2ee.com/en/car/changan/benni/6286668
27	1000000	Silver	96500	Opel	Astra	Alexandria	2/16/2024	0	0	0	0	https://eg.hatla2ee.com/en/car/opel/astra/6286677

Approach

- *After removing extreme outliers, roughly 100k rows were removed.*



- *Observing correlations between price and different features.*

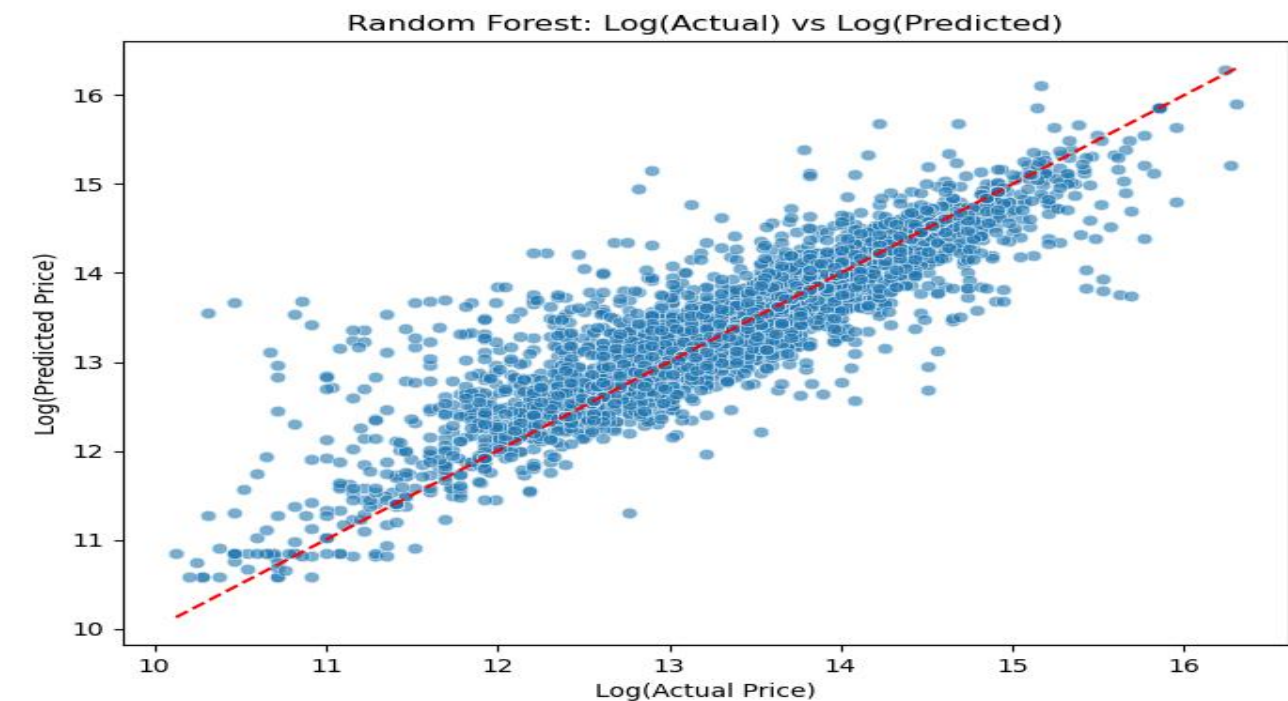
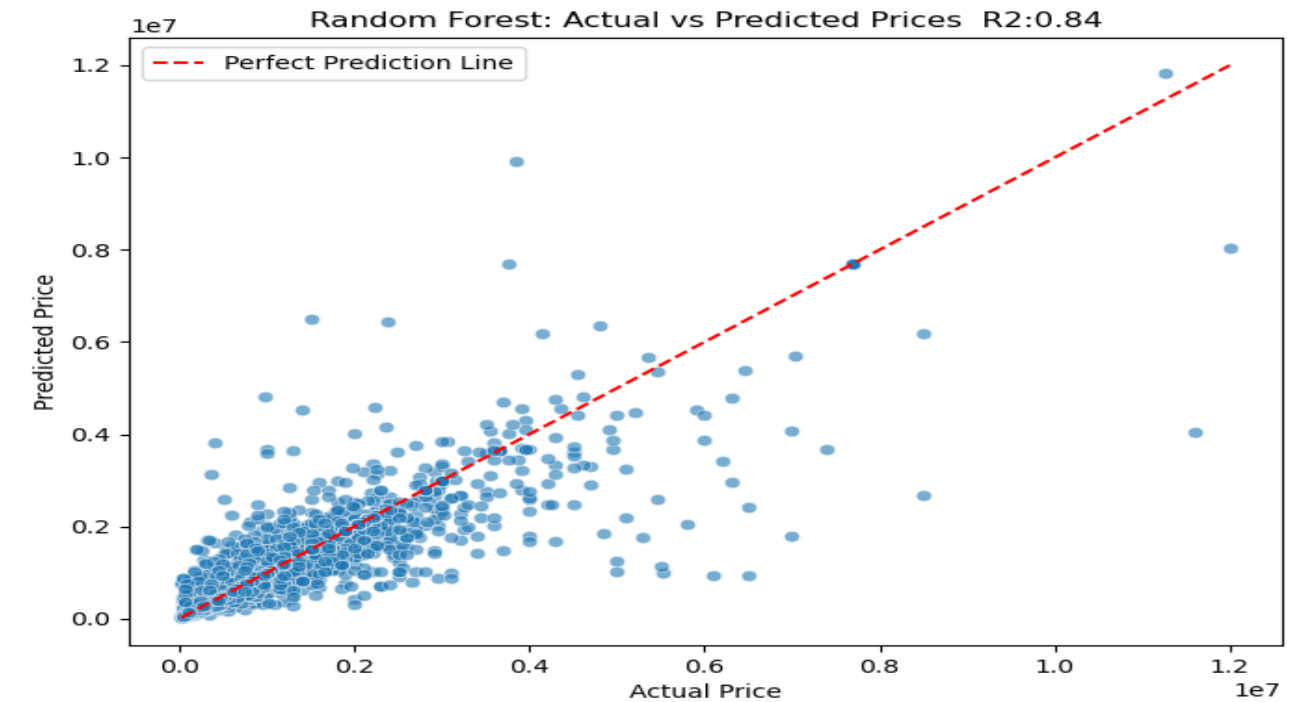


Approach

- *After encoding the brand and model columns using TargetEncoder, the next step was training models. Models tested were Linear Regression, Decision Tree and Random Forest.*
- *Best performing model was the Random Forest, with an R^2 of 84, MAE of 153000 and RMSE of 402000 (approximately).*
- *A decent R^2 score. As for the MAE and the RMSE, the scores are relatively high but in this case it's completely normal since car prices are highly skewed, especially with expensive and luxurious brands, prices take a big turn and they're harder to predict.*

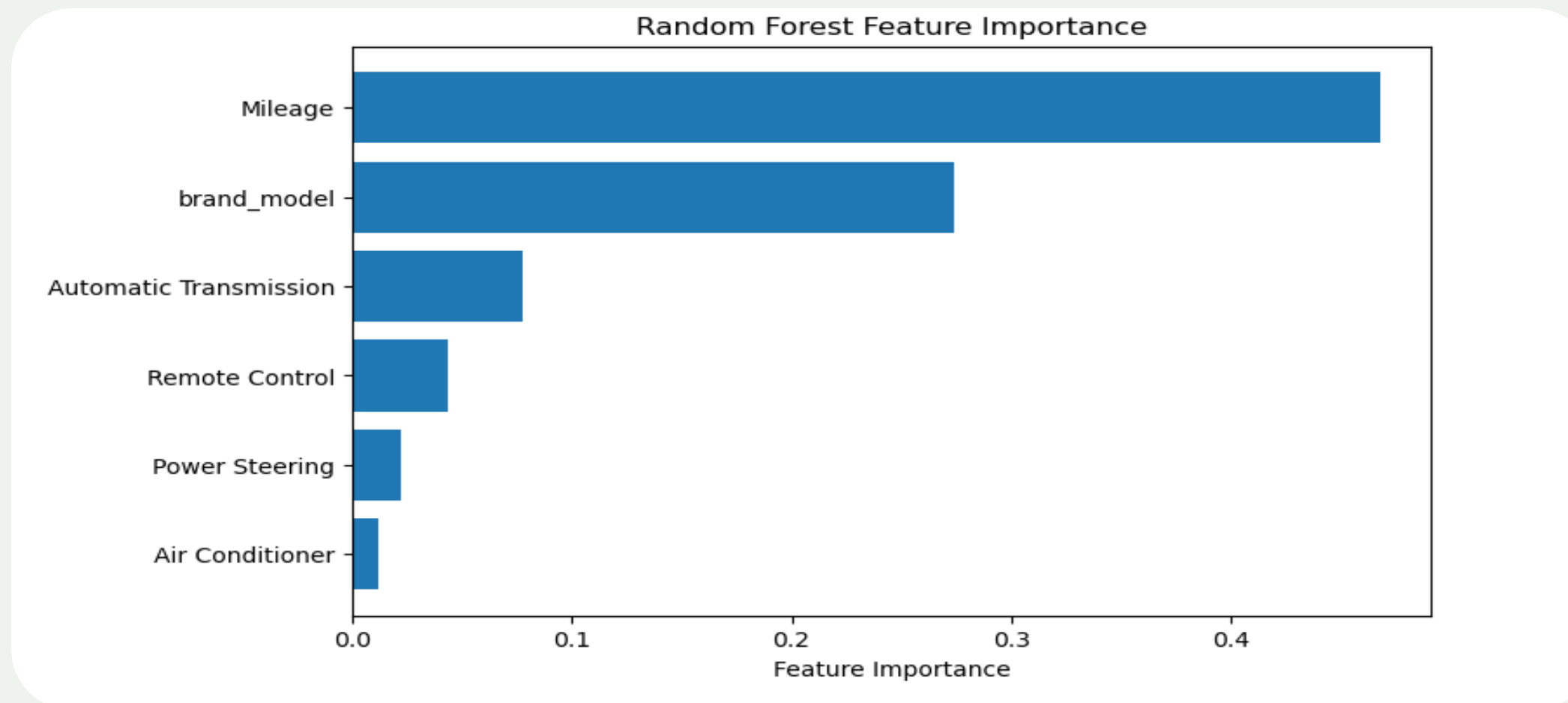
Graphs

- *After plotting the Actual and predicted prices for the random forest model, most points are centered around the perfect line, but some are scattered the higher the price.*
- *For visual clarity, I log transformed the prices to reduce visual skewness and to see how the model actually performed.*



Graphs

- *The feature importance chart highlights which variables had the strongest influence on predicting used car prices. Mileage emerged as the most significant factor. The brand and model combination also played a major role, reflecting how brand reputation and model popularity affect pricing.*



Interactive Interface



1	Column1	Name	Price
67	69	Fiat 500C 2015	800000

Used Car Price Prediction App

Enter car details below to estimate its price.

Mileage (in km):

100000

-

+

Automatic Transmission:

No

Air Conditioner:

No

Power Steering:

No

Remote Control:

No

Brand and Model (e.g., Toyota Corolla):

Fiat 500C

Predict Price

Estimated Price: 797,191 EGP

1	Column1	Name	Price
90	92	Peugeot 301 2014	550000

Used Car Price Prediction App

Enter car details below to estimate its price.

Mileage (in km):

137000

-

+

Automatic Transmission:

Yes

Air Conditioner:

Yes

Power Steering:

Yes

Remote Control:

Yes

Brand and Model (e.g., Toyota Corolla):

Peugeot 301

Predict Price

Estimated Price: 533,133 EGP

1	Column1	Name	Price
130	135	Opel Astra 2021	1250000

Used Car Price Prediction App

Enter car details below to estimate its price.

Mileage (in km):

15000

-

+

Automatic Transmission:

Yes

Air Conditioner:

Yes

Power Steering:

Yes

Remote Control:

Yes

Brand and Model (e.g., Toyota Corolla):

Opel Astra

Predict Price

Estimated Price: 1,116,531 EGP

Conclusion

- - *The Random Forest model achieved strong predictive accuracy.*
- - *The model's performance degrades the higher the prices get, since car prices are highly skewed, especially luxurious ones.*
- *No data leakage since the corss-validation achieved similar scores.*
- *Mileage and brand are the most important features that affect the price of a used car, other features doesn't really affect the price that much.*

Thank you!



For technical part:

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